

Rahul Rajesh

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EDUCATION

B.S. in Math (Statistics), B.S. in Computer Science (Artificial Intelligence), Minor in Studio Art **GPA: 4.0**
Barrett, The Honors College, Arizona State University *Expected Graduation: May 2028*
Honors/Awards: 2nd @ Strategy x Devlabs Hackathon (50+ Teams), Dean's List (4/4), President Scholarship, Tillman Scholar
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Discrete Mathematics, Software Engineering, Probability, Applied Statistics, Linear Algebra, Digital Design, Vector/Advanced Calculus, Applied Regression

SKILLS

Languages: Java, Python, C, C++, SQL, R
Frameworks and Backend: Spring Boot, Spring Data JPA, Hibernate, Docker, JavaFX, Maven, Postman
AI/ML & RAG: Pandas, NumPy, Vector Search (pgvector), Embeddings (BGE), Reranking, LangGraph, Reinforcement Learning (MuJoCo, Isaac Sim), Computer Vision (YOLO/Ultralytics)
Infra & Data: PostgreSQL, BigQuery, Pub/Sub, Tableau
Other: MATLAB, HTML/CSS, ROS2

PROJECTS

Simple Banking System (SpringBoot Rest API) **Tempe, AZ**
Solo Project *Feb 2026*

- Built a layered Spring Boot RESTful API (Controller-Service-Repository) using Spring Data JPA and Hibernate for ORM, MySQL integration with HikariCP connection, DTO mapping, and exception handling; tested endpoints with Postman

ReThread - AI-Powered Sustainable Shopping Extension (2nd Place, DevHacks x Strategy) **Tempe, AZ**
Team Lead *Mar 2025*

- Developed a Chrome extension that integrates computer vision and keyword matching to identify fashion products and surface sustainable resale alternatives in real time from 2nd hand websites (Poshmark) to promote textile recycling
- Utilized Google Cloud Vision and third-party APIs to perform image recognition and map features to structured metadata.

Campus Compass - Campus Life Involvement Agent (ScaleU + Principled Innovation Academy) **Tempe, AZ**
Team Member *Mar 2025*

- Developed an AI agent chatbot using Python and AWS Bedrock to help students discover personalized campus activities and organizations based on their interests, major, and goals to support their adjustment to college life
- Scraped data from ASU's SunDevilSync platform to populate the chatbot's knowledge base hosted on an S3 bucket

EXPERIENCE

EssilorLuxottica **New York City, NY**
Data Analytics + AI Intern *May 2026 - Jul 2026*

- Built an automated ETL pipeline in R that ingests raw SAP and Hyperion extracts (Excel/CSV, inconsistent schemas and multi-row headers), cleans and joins them via automated matching logic, and outputs a single unified dataset powering a live Tableau dashboard — replacing a fully manual process and cutting turnaround from ~4 hours to ~15 minutes per cycle.
- Contributed to EssilorLuxottica's internal conversational AI platform (RAG-based analytics assistant) — built and containerized an embedding pipeline that generates vectors via the BGE model from BigQuery source data and writes them to a pgvector-backed PostgreSQL store, and implemented a BGE reranker to re-score top-K vector search results before they reach the LLM, improving retrieval relevance on the team's internal eval set.

Robotics Actuators and Dynamics Research Lab (Prof. Jiefeng Sun) **Tempe, AZ**
Simulations and Reinforcement Learning Lead *Jul 2025 – Present*

- Developing ROS2-based velocity estimation pipeline for a biomimetic quadruped robot, fusing IMU and OptiTrack motion capture with a Kalman filter to reduce sensor noise; built initial ROS2 nodes and now implementing Kalman algorithms
- Training YOLO/Ultralytics computer vision models to detect and classify invasive Diamondback Moths (adult and larval stages) in botanical shipments and containers for a border biosecurity screening system
- Previously built high-fidelity IsaacSim and MuJoCo simulations of a high-DOF spined-quadruped robot, including joint dynamics, actuator models, RL-ready environments to support locomotion, robust maneuvers, and stability

Arizona State University D1 Women's Soccer **Tempe, AZ**
Student Manager + Statistical Analyst *Dec 2025 – May 2026*

- Built Python-based data (Pandas, NumPy) pipelines to clean, and normalize multi-source player tracking and event-level datasets across seasons and competitions for tactical scouting insights; implemented aggregation logic to compute availability metrics and percentile-based summaries at player and team levels to support roster evaluation and performance

Solar Devils (1st Place, ASU Change the World 2024) **Tempe, AZ**
Integrations Team Member *Aug 2024 – May 2025*

- Built a SQLite3 & Python data pipeline to ingest and query the vehicle's CAN bus telemetry network to support data integration efforts to support a solar-powered vehicle to enhance performance through structured data storage and retrieval
- Enabled real-time performance analysis and troubleshooting by exposing vehicle data to on-board visualization systems